Applied Statistics (MATH10096)

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Inference with GLMs

GLM fitting

- Recall that a GLM models an $n \times 1$ vector of independent response variables $\mathbf{y}$, where $\boldsymbol{\mu} \equiv \mathbb{E}(\mathbf{y})$, via
$$g(\mu_i) = \mathbf{x}_i^{\mathsf{T}} \boldsymbol{\beta}, \quad \text{and} \quad y_i \sim f(y_i; \theta_i, \phi) \quad i = 1, \ldots, n.$$
- Here:
  - $f(y_i; \theta_i, \phi)$ denotes an exponential family distribution with canonical parameter θ_i.
  - θ_i is determined by μ_i via $\mu_i = b'(\theta_i)$ and, ultimately, by $\boldsymbol{\beta}$.
- Because the response variables are mutually independent, the likelihood of $\boldsymbol{\beta}$ is

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} f(y_i; \theta_i, \phi).$$

- Hence, the log likelihood of $\boldsymbol{\beta}$ is

$$\log L(\boldsymbol{\beta}) = l(\boldsymbol{\beta}) = \sum_{i=1}^{n} l_i(\boldsymbol{\beta}) = \sum_{i=1}^{n} \log f(y_i; \theta_i, \phi) = \sum_{i=1}^{n} \left\{ \frac{\omega_i [y_i \theta_i - b(\theta_i)]}{\phi} + c(y_i, \phi) \right\}.$$

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- Finding the maximum likelihood estimate of $\boldsymbol{\beta}$, which, as before, we assume to be a $p \times 1$ vector, requires the score vector, i.e., the vector formed by the partial derivatives of $l(\boldsymbol{\beta})$ with respect to each β_j, for $j = 1, \ldots, p$, that is,

$$\mathbf{U}(\boldsymbol{\beta}) = \begin{pmatrix} \frac{\partial l(\boldsymbol{\beta})}{\partial \beta_1} \\ \vdots \\ \frac{\partial l(\boldsymbol{\beta})}{\partial \beta_p}. \end{pmatrix}$$

- We have that

$$\frac{\partial l(\boldsymbol{\beta})}{\partial \beta_j} = \frac{\partial}{\partial \beta_j} \sum_{i=1}^{n} l_i(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta}), \qquad j = 1, \ldots, p.$$

- To differentiate the log likelihood, we apply the chain rule:

$$\frac{\partial l_i(\boldsymbol{\beta})}{\partial \beta_j} = \frac{\partial l_i}{\partial \theta_i} \frac{\mathrm{d}\theta_i}{\mathrm{d}\mu_i} \frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i} \frac{\partial \eta_i}{\partial \beta_j}.$$

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$\hookrightarrow$ Let us now workout each derivative on the right-hand side of the equality obtain when applying the chain rule. We have

$$
\begin{aligned}
\frac{\partial l_i}{\partial \theta_i} &= \frac{\partial}{\theta_i}\left\{\frac{\omega_i}{\phi}[y_i\theta_i - b(\theta_i)] + c(y_i, \phi)\right\} \\
&= \frac{\omega_i}{\phi}[y_i - b'(\theta_i)] \\
&= \frac{\omega_i}{\phi}(y_i - \mu_i), \quad \mu_i = \mathbb{E}(y_i) = b'(\theta_i).
\end{aligned}
$$

$\hookrightarrow$ For determining $\frac{\mathrm{d}\theta_i}{\mathrm{d}\mu_i}$, we note that $\mu_i = b'(\theta_i)$, and so

$$\frac{\mathrm{d}\mu_i}{\mathrm{d}\theta_i} = b''(\theta_i),$$

and because b' is a one to one function, one must have

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}\mu_i} = \frac{1}{\frac{\mathrm{d}\mu_i}{\mathrm{d}\theta_i}} = \frac{1}{b''(\theta_i)}.$$

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- Similarly, and noting that $g(\mu_i) = \eta_i$, we have that

$$\frac{\mathrm{d}\eta_i}{\mathrm{d}\mu_i} = g'(\mu_i),$$

and because g is a smooth monotonic function, then

$$\frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i} = \frac{1}{\frac{\mathrm{d}\eta_i}{\mathrm{d}\mu_i}} = \frac{1}{g'(\mu_i)}.$$

- Finally, note that

$$\eta_i = \mathbf{x}_i^\mathrm{T}\boldsymbol{\beta} = \sum_{j=1}^{p} \beta_j x_{ij} \Rightarrow \frac{\partial \eta_i}{\partial \beta_j} = x_{ij},$$

- Putting all the terms together leads to

$$\frac{\partial l_i(\boldsymbol{\beta})}{\partial \beta_j} = \frac{\omega_i}{\phi}(y_i - \mu_i) \times \frac{1}{b''(\theta_i)} \times \frac{1}{g'(\mu_i)} \times x_{ij},$$

↪ We thus have

$$\begin{aligned}\frac{\partial}{\partial \beta_j} l(\boldsymbol{\beta}) &= \frac{1}{\phi}\sum_{i=1}^{n} \frac{(y_i-\mu_i)x_{ij}}{\frac{b''(\theta_i)}{\omega_i} g'(\mu_i)} \\ &= \frac{1}{\phi}\sum_{i=1}^{n} \frac{(y_i-\mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}, \quad j = 1,\ldots,p.\end{aligned}$$

↪ To maximize $l(\boldsymbol{\beta})$, we find the vector $\hat{\boldsymbol{\beta}}$ for which the vector of derivatives is equal to $\mathbf{0}$.

↪ Note that $\hat{\boldsymbol{\beta}}$ will not depend on ϕ.

↪ These equations do not have an analytic solution, so we resort to a numerical procedure. To do this, we require the matrix of second derivatives of the log likelihood.

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↪ Let us find a generic entry of the matrix of second derivatives of the log likelihood:

$$\begin{aligned}\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \beta_k \beta_j} &= \frac{\partial}{\partial \beta_k}\left(\frac{\partial l(\boldsymbol{\beta})}{\partial \beta_j}\right) \\ &= \frac{\partial}{\partial \beta_k}\left(\sum_{i=1}^{n} \frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta})\right) \\ &= \sum_{i=1}^{n} \frac{\partial}{\partial \beta_k}\left(\frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta})\right).\end{aligned}$$

↪ We have that

$$\frac{\partial}{\partial \beta_k}\left(\frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta})\right) = \frac{\partial}{\partial \beta_k}\left(\frac{1}{\phi}\frac{(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}\right) = \frac{1}{\phi}\frac{\partial}{\partial \beta_k}\left(\frac{(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}\right).$$

- To calculate
$$\frac{\partial}{\partial \beta_k}\left(\frac{(y_i-\mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}\right),$$
we apply the chain rule again as follows:
$$\frac{\mathrm{d}\left(\frac{(y_i-\mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}\right)}{\mathrm{d}\mu_i}\frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i}\frac{\partial\eta_i}{\partial\beta_k}.$$
- As before,
$$\frac{\mathrm{d}\mu_i}{\mathrm{d}\eta_i}=\frac{1}{g'(\mu_i)},\quad \text{and}\quad \frac{\partial\eta_i}{\partial\beta_k}=x_{ik}.$$
- For the first term we have that

$$\frac{\mathrm{d}\left(\frac{(y_i-\mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}\right)}{\mathrm{d}\mu_i}=\frac{\mathrm{d}}{\mathrm{d}\mu_i}\left((y_i-\mu_i)x_{ij}\times\frac{1}{V(\mu_i)g'(\mu_i)}\right)=-x_{ij}\frac{1}{V(\mu_i)g'(\mu_i)}+\frac{\mathrm{d}}{\mathrm{d}\mu_i}\left(\frac{1}{V(\mu_i)g'(\mu_i)}\right)(y_i-\mu_i)x_{ij}.$$

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- We further have that

$$\frac{\mathrm{d}}{\mathrm{d}\mu_i}\left(\frac{1}{V(\mu_i)g'(\mu_i)}\right) = -\frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{g'(\mu_i)^2 V(\mu_i)^2}.$$

- Therefore,

$$\frac{\mathrm{d}\left(\frac{(y_i-\mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)}\right)}{\mathrm{d}\mu_i} = -x_{ij}\frac{1}{V(\mu_i)g'(\mu_i)} - \frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{g'(\mu_i)^2 V(\mu_i)^2}(y_i-\mu_i)x_{ij}.$$

- Putting all the terms together leads to

$$\begin{aligned}\frac{\partial^2 l(\boldsymbol{\beta})}{\partial\beta_k\beta_j} &= -\frac{1}{\phi}\sum_{i=1}^{n}\left\{\left[x_{ij}\frac{1}{V(\mu_i)g'(\mu_i)} + \frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{g'(\mu_i)^2 V(\mu_i)^2}(y_i-\mu_i)x_{ij}\right]\frac{1}{g'(\mu_i)}x_{ik}\right\}\\ &= -\frac{1}{\phi}\sum_{i=1}^{n}\left\{\frac{x_{ik}x_{ij}}{g'(\mu_i)^2 V(\mu_i)} + \frac{(y_i-\mu_i)x_{ij}x_{ik}}{g'(\mu_i)^2 V(\mu_i)}\left(\frac{g''(\mu_i)}{g'(\mu_i)} + \frac{V'(\mu_i)}{V(\mu_i)}\right)\right\}.\end{aligned}$$

- Note that

$$\mathbb{E}\left(\frac{\partial^2 l}{\partial\beta_k\partial\beta_j}\right) = -\frac{1}{\phi}\sum_{i=1}^{n}\frac{x_{ik}x_{ij}}{g'(\mu_i)^2 V(\mu_i)},$$

- Defining $\mathrm{w}_i^{-1} = g'(\mu_i)^2 V(\mu_i)$ and $\mathrm{W} = \mathrm{diag}(\mathrm{w}_1, \ldots, \mathrm{w}_n)$, we therefore have that

$$\boldsymbol{\mathcal{I}} = -\mathrm{E}(\partial^2 l/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\mathsf{T}) = \mathbf{X}^\mathsf{T}\mathbf{W}\mathbf{X}/\phi.$$

- From the general properties of maximum likelihood estimators we have that, in the large-sample limit,

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \boldsymbol{\mathcal{I}}^{-1}), \quad \boldsymbol{\mathcal{I}} = \mathbf{X}^\mathsf{T}\mathbf{W}\mathbf{X}/\phi.$$

- For distributions with known scale parameter ϕ (e.g., the Poisson distribution) this result can be used to directly find confidence intervals for the parameters.

- If the scale parameter is unknown (as it is the case for, e.g., the normal and gamma distributions), then it must be estimated and intervals must be based on an appropriate t distribution.

Inference with GLMs

GLM fitting

- Applying Newton's method to find $\hat{\beta}$ amounts to iterating (from some initial guess $\boldsymbol{\beta}^{(0)}$)

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \mathbf{H}(\boldsymbol{\beta}^{(t)})^{-1}\mathbf{U}(\boldsymbol{\beta}^{(t)}), \quad \mathbf{H}(\boldsymbol{\beta}) = \frac{\partial^2}{\partial \boldsymbol{\beta}\boldsymbol{\beta}^{\mathsf{T}}} l(\boldsymbol{\beta}),$$

  until successive $\boldsymbol{\beta}^{(t)}$ do not change.

- A useful feature of Newton's method is that it converges just as well with the second derivative matrix replaced by its expectation.

- This modification is known as the Fisher scoring method, whose updating equations are as follows:

$$\begin{aligned}\boldsymbol{\beta}^{(t+1)} &= \boldsymbol{\beta}^{(t)} - (\mathbb{E}[\mathbf{H}(\boldsymbol{\beta}^{(t)})])^{-1}\mathbf{U}(\boldsymbol{\beta}^{(t)}) \\ &= \boldsymbol{\beta}^{(t)} + \boldsymbol{\mathcal{I}}(\boldsymbol{\beta}^{(t)})^{-1}\mathbf{U}(\boldsymbol{\beta}^{(t)})\end{aligned}$$

- Note that

$$\begin{aligned}\boldsymbol{\mathcal{I}}(\boldsymbol{\beta}^{(t)})^{-1} &= (\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{X})^{-1}\phi \\ \mathbf{U}(\boldsymbol{\beta}^{(t)}) &= \frac{1}{\phi}\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{G}^{(t)}(\mathbf{y} - \boldsymbol{\mu}^{(t)}), \quad \mathbf{G} = \mathrm{diag}\{g'(\mu_1), \ldots, g'(\mu_n)\}.\end{aligned}$$

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↪ Hence the updating equations of the Fisher scoring algorithm are given by

$$\begin{aligned}\boldsymbol{\beta}^{(t+1)} &= \boldsymbol{\beta}^{(t)} + (\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{G}^{(t)}(\mathbf{y}-\boldsymbol{\mu}^{(t)})\\ &= (\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\{\mathbf{G}^{(t)}(\mathbf{y}-\boldsymbol{\mu}^{(t)})+\mathbf{X}\boldsymbol{\beta}^{(t)}\}\\ &= (\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{W}^{(t)}\mathbf{z}^{(t)}\end{aligned}$$

↪ In the last line, $z_i^{(t)} = g'(\mu_i^{(t)})(y_i - \mu_i^{(t)}) + \eta_i^{(t)}$ and $\eta_i^{(t)} = \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}^{(t)}$.

↪ So $\boldsymbol{\beta}^{(t+1)}$ is recognisable as the minimiser of the weighted sum of squares

$$\sum_i w_i(z_i - \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})^2.$$

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GLM fitting

↪ Hence maximum likelihood estimation for a GLM is achieved using the iteratively re-weighted least squares (IRLS) algorithm.

1. Set $t = 0$, $\mu_i^{(0)} = y_i + \Delta_i$ and $\eta_i^{(0)} = g(\mu_i^{(0)})$, where $\Delta_i = 0$ usually, but may be a small constant if needed to ensure that $\eta_i^{(0)}$ is well defined and finite (for instance for count data, Poisson regression with a log link, if some counts are zero). Then iterate the following two steps until convergence.

2. Form $z_i^{(t)} = g'(\mu_i^{(t)})(y_i - \mu_i^{(t)}) + \eta_i^{(t)}$ and $w_i^{(t)} = \{g'(\mu_i^{(t)})^2 V(\mu_i^{(t)})\}^{-1}$.

3. Find $\beta^{(t+1)} = \arg\min_\beta \sum_i w_i^{(t)}(z_i^{(t)} - \mathbf{x}_i^\top \beta)^2$, by standard linear model methods. Increment t.

Inference with GLMs

GLM fitting: a brief aside on weighted least squares

↪ Recall that an expression for the value of $\boldsymbol{\beta}$ minimizing

$$\sum_{i=1}^{n}(z_i - \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})^2,$$

is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{z}.$$

↪ Let us verify that $\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{z}$ is indeed the minimizer of the weighted sum of squares

$$\sum_{i=1}^{n} w_i(z_i - \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})^2.$$

↪ Here $\mathbf{W}$ is an $n \times n$ diagonal matrix such that $W_{ii} = \mathrm{w}_i$.

Inference with GLMs

GLM fitting: a brief aside on weighted least squares

- To this end let $\tilde{\mathbf{W}}$ be the diagonal matrix such that $\tilde{W}_{ii} = \sqrt{w_i}$. It is clear that $\mathbf{W} = \tilde{\mathbf{W}}\tilde{\mathbf{W}}$.
- We have

$$\begin{aligned}\sum_{i=1}^{n} w_i(z_i - \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})^2 &= \sum_{i=1}^{n}\{\sqrt{w_i}(z_i - \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})\}^2 \\ &= \sum_{i=1}^{n}(\sqrt{w_i}z_i - \sqrt{w_i}\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})^2 \\ &= \sum_{i=1}^{n}(\tilde{z}_i - \tilde{\mathbf{x}}_i^{\mathsf{T}}\boldsymbol{\beta})^2.\end{aligned}$$

- Here $\tilde{\mathbf{z}} = \tilde{\mathbf{W}}\mathbf{z}$ and $\tilde{\mathbf{X}} = \tilde{\mathbf{W}}\mathbf{X}$.

$\hookrightarrow$ So using the formal expression for a least squares estimate of β gives

$$\begin{aligned}\hat{\beta} &= (\tilde{\mathbf{X}}^{\mathsf{T}}\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^{\mathsf{T}}\tilde{\mathbf{z}} \\ &= (\mathbf{X}^{\mathsf{T}}\tilde{\mathbf{W}}\tilde{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\tilde{\mathbf{W}}\tilde{\mathbf{W}}\mathbf{z} \\ &= (\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{z}.\end{aligned}$$

- Recall that after expressing the model in a GLM form, we have

$$\log(\mu_i) = \beta_1 + \beta_2 t_i, \qquad i = 1, \ldots, 13.$$

- As a starting point it makes sense to try a Poisson distribution with a log link (*Poisson regression*).

- To implement the IRLS algorithm we need $g'(\mu) = (\log \mu)' = \frac{1}{\mu}$.

- We further have that for the Poisson distribution $V(\mu) = \mu$.

- Note that at any iteration t we have

$$z_i^{(t)} = g'(\mu_i^{(t)})(y_i - \mu_i^{(t)}) + \eta_i^{(t)} = \frac{1}{\mu_i^{(t)}}(y_i - \mu_i^{(t)}) + \eta_i^{(t)} = \frac{y_i}{\mu_i^{(t)}} - 1 + \eta_i^{(t)},$$

$$\omega_i^{(t)} = \{g'(\mu_i^{(t)})^2 V(\mu_i^{(t)})\}^{-1} = \left(\frac{1}{\mu_i^{(t)2}} \mu_i^{(t)}\right)^{-1} = \mu_i^{(t)}.$$

Inference with GLMs

GLM fitting: AIDS example

```r
y <- c(12,14,33,50,67,74,123,141,165,204,253,246,240)
t <- 1:13
mu <- y; eta <- log(mu) ## initialize mean and linear predictor
converged <- FALSE
while (!converged) {
  z <- y/mu - 1 + eta ## create pseudodata
  w <- mu             ## create iterative weights
  fit <- lm(z ~ t, weights = w) ## fit working model
  eta.old <- eta      ## store old linear predictor
  eta <- fitted(fit)  ## get new linear predictor (X \beta)
  mu <- exp(eta)      ## get new mean
  ## test for convergence ...
  if (max(abs(eta-eta.old))<1e-6*mean(abs(eta))) converged <- TRUE
}
coef(fit)

## (Intercept)           t
##   3.1405895   0.2021212
```

Inference with GLMs

GLM fitting: AIDS example

- Note that we have used the `lm` function with an argument that is new to us: `weights`. By typing `help(lm)`, we find the following description for this argument:

"an optional vector of weights to be used in the fitting process. (...) If non-NULL, weighted least squares is used with weights weights (that is, minimizing sum$(w * e^2)$) (...)"

- This is exactly what we want!

Inference with GLMs

GLM fitting: AIDS example

↪ It is also easy to implement the IRWLS without using the `lm` function.

```
mu <- y; eta <- log(mu)
t_vec <-cbind(rep(1, length(t)), t)
converged <- FALSE
iter <- 0 #counts how many iterations until convergence
while (!converged) {
  z <- y/mu - 1 + eta
  w <- as.vector(mu)
  coefs <- solve(t(t_vec)%*%diag(w)%*%t_vec)%*%t(t_vec)%*%diag(w)%*%cbind(z)
  eta.old <- eta
  eta <- t_vec%*%coefs
  mu <- exp(eta)
  if (max(abs(eta-eta.old))<1e-6*mean(abs(eta))) converged <- TRUE
  iter <- iter + 1
}
as.numeric(coefs)
## [1] 3.1405895 0.2021212
iter
## [1] 4
```

Inference with GLMs

GLM fitting: AIDS example

↪ We now compare with the results obtained using the `glm` function.

```
mod1 <- glm(y ~ t, family = poisson)
summary(mod1)

##
## Call:
## glm(formula = y ~ t, family = poisson)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) 3.140590   0.078247   40.14   <2e-16 ***
## t           0.202121   0.007771   26.01   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 872.206  on 12  degrees of freedom
## Residual deviance:  80.686  on 11  degrees of freedom
## AIC: 166.37
##
## Number of Fisher Scoring iterations: 4
```